

Heart

- Heart is a part of circulatory system
- Double circulatory system is found in our body
 - ↳ 1628
 - ↳ William Harvey

Function:-

- ↳ Blood Pumping
- ↳ size $\rightarrow$ Fist (थंडी)

Location.

mid of the chest

In children $\rightarrow$ ~~250-350gm~~ ^{250-380gm}

New born $\rightarrow$ 20-25 gm

Adults $\rightarrow$ 230-250 gm

Boys $\rightarrow$ 250-280 gm

Structure

- 4-chamber
- Right Atrium
- Left Atrium
- Right ventricle
- Left ventricle.

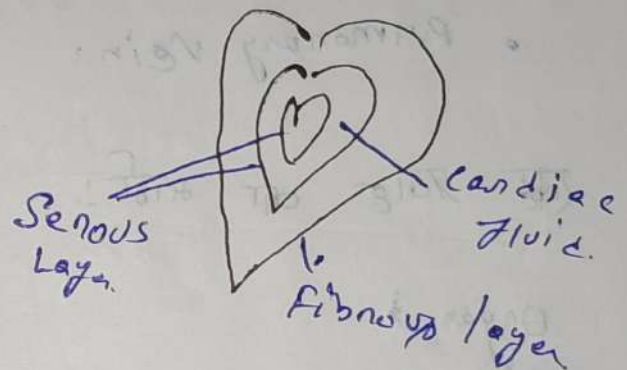


Heart structure :

- Semi-lunar valve
- Pulmonary valve
- Aorta valve
- Tricuspid valve
- Bi-cuspid valve

Layers

- Fibrous layer
- Serous layer
- Pericardium layer
= Fibrous + serous



Location

- Media sternum space (मध्यस्थ)
- Cardiac Notch
- फोफेरा (R)
- Media sternum space
- मध्यस्थ स्टेनम स्पेस
- Cardiac Notch (हृदय गुहा)

Function :-

• Blood Pumping

• Heart $\rightarrow$ RBC $\rightarrow$ CO_2 + O_2

Alveoli

• Valve $\rightarrow$ Back flow

• Organs

• Lungs

• Superior vena cava:

• Inferior vena cava:

• Pulmonary vein:

रक्त वाहक का मार्ग :-

Organ $\rightarrow$

Blood $\rightarrow$

S.V.C + I.V.C $\rightarrow$

R.A. $\rightarrow$

T.C.V. $\rightarrow$

R.V. $\rightarrow$

P.A. $\rightarrow$ Lungs $\rightarrow$ L.A. $\rightarrow$ L.V. $\rightarrow$ Aorta Valve $\rightarrow$

Organ

• Natural pacemaker

• $\rightarrow$ Impulse + Na^+ , mg^{++} , K^+

• SA (sino Atricle node)

• A.V node (Atriculo Ventricular node)

• Bundle of His

• Purkinje fibre

• Li-Battery $\rightarrow$ 15-20 gm

• Light metal

• 20 gm (max)

• Heart beat

$\hookrightarrow$ sound

$\hookrightarrow$ Lub - dub.

$\hookrightarrow$ Murn Jerny

$\hookrightarrow$ contraction + relaxation.

1 systole + 1 diastole = heart beat

• 3 sec + 5 sec = 8 sec

Heart beat graph

- Human $\rightarrow 72/\text{min}$
- Embryo $\rightarrow 200/\text{min}$
- Children $\rightarrow 130-150/\text{min}$
- Elephant + Whale $\rightarrow 20-25/\text{min}$
- Mouse/Rat $\rightarrow 250/\text{min}$
- Exercise $\rightarrow 90-95/\text{min}$
- Athlete $\rightarrow 60-65/\text{min}$

60-65 b/min (Bradycardia)

130-150 b/min (Tachycardia)

1 heart beat = 70 ml

72 min $\therefore 70 \times 72$

$= 5040 \text{ ml}$

$= 5 \text{ litre.}$

↳ Artery $\rightarrow$ Coronary Artery

Heart Attack

- ↳ Blood circulation
- ↳ Cholesterol.

-HDL

LDL

↳ B.P. (रक्तचाप उच्च - नीचे)

↳ O₂

↳ फ्लो स्लो

Heart failure

Lack of O₂

Cardiac arrest

Valve block

Back flow

By Pass surgery

- Artificial Artery
- (obstruction/blockage)

Does heart Relax

- No [According to NTPC]

But in reality. yes.

LS 15 + 10

.3 .5

.5 - .3 = .2 sec it relaxes

B.P. $\rightarrow$ Gl and

$\hookrightarrow$ Adrenal

$\hookrightarrow$ medulla oblongata

Heart Transplant

$\hookrightarrow$ Capetown (S.A.)

$\hookrightarrow$ Christiaan Barnard.

In India : 1994

$\hookrightarrow$ P. Venugopal

Heart Hole



Chamber

$\hookrightarrow$ fish (A'ces) $\rightarrow$ 2c (1A, 1V)

$\hookrightarrow$ Amphibian $\rightarrow$ 3c (2A, 1V)

$\hookrightarrow$ Reptiles $\rightarrow$ 3c

$\hookrightarrow$ अणुद्वि (द्विचकली, मीरिह-उड)

$\hookrightarrow$ Mammals $\rightarrow$ 4c (2A, 2V)

$\hookrightarrow$ birds $\rightarrow$ 4c (2A, 2V)